

BAAZ JKBOSE

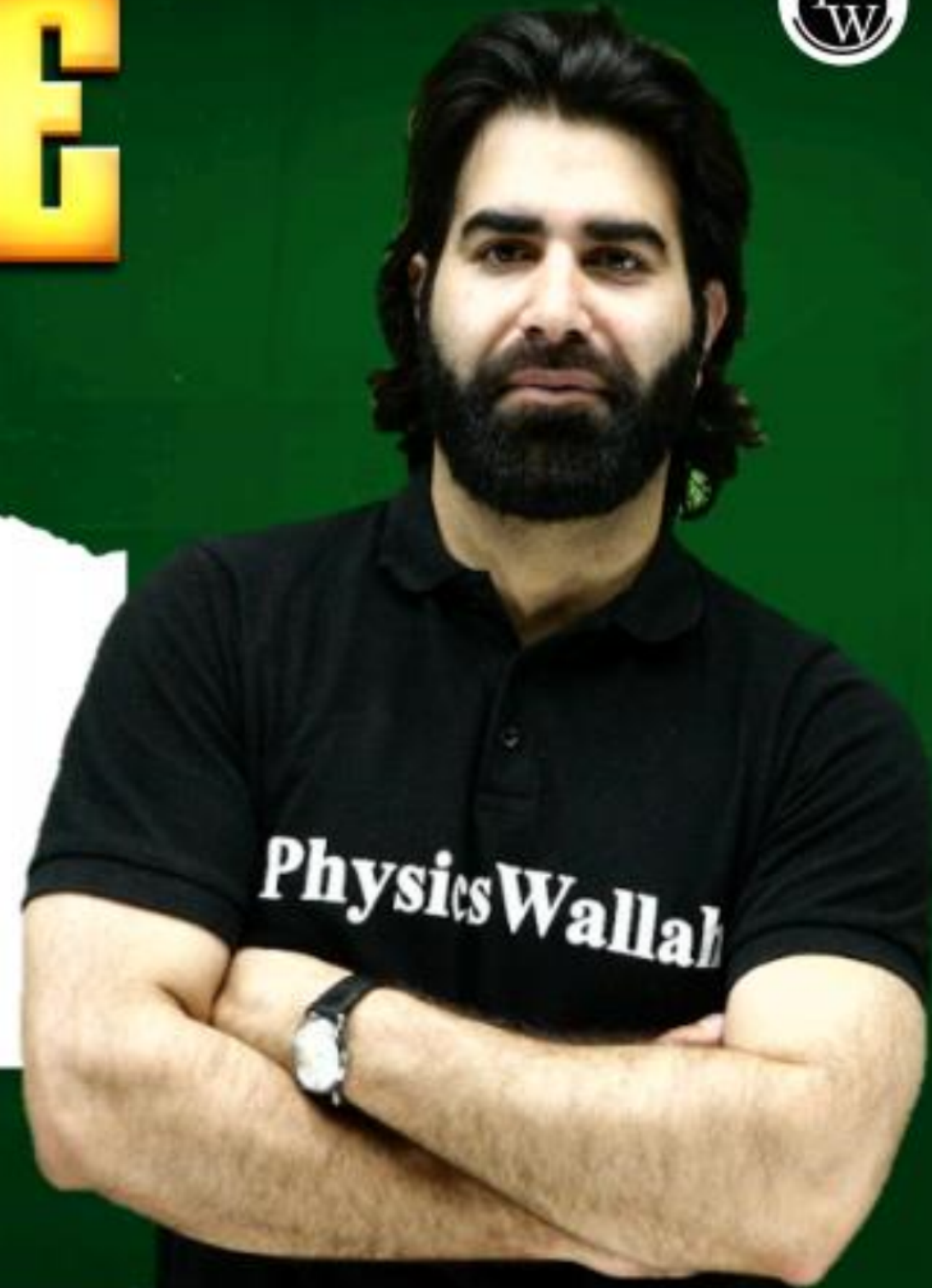
2026 CLASS-11TH

Locomotion and
Movement

LECTURE-2

Zoology

Dr Adil Altaf sir



Topics To Be Covered

1

Muscle Fibre, Muscle Contraction

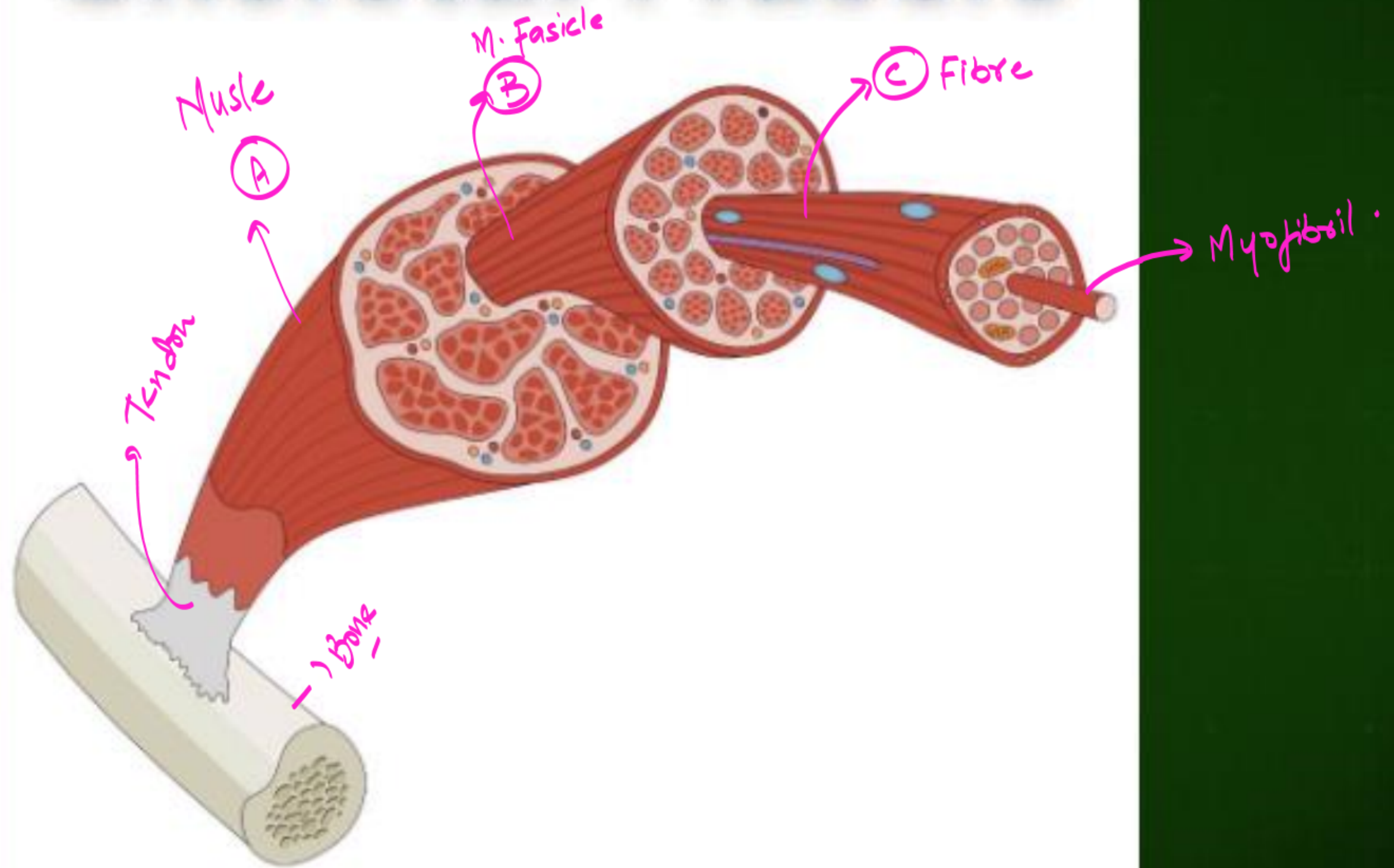
Revision

→ e.g of Amoeboid Movement = ①

- Muscle = Involuntary
Non-Straited = ②

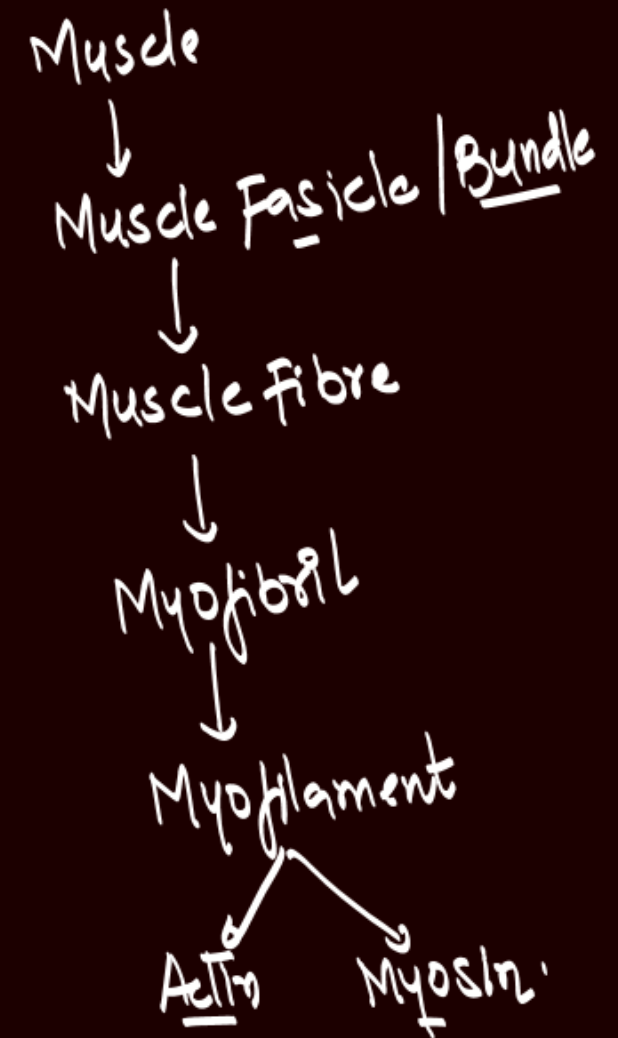
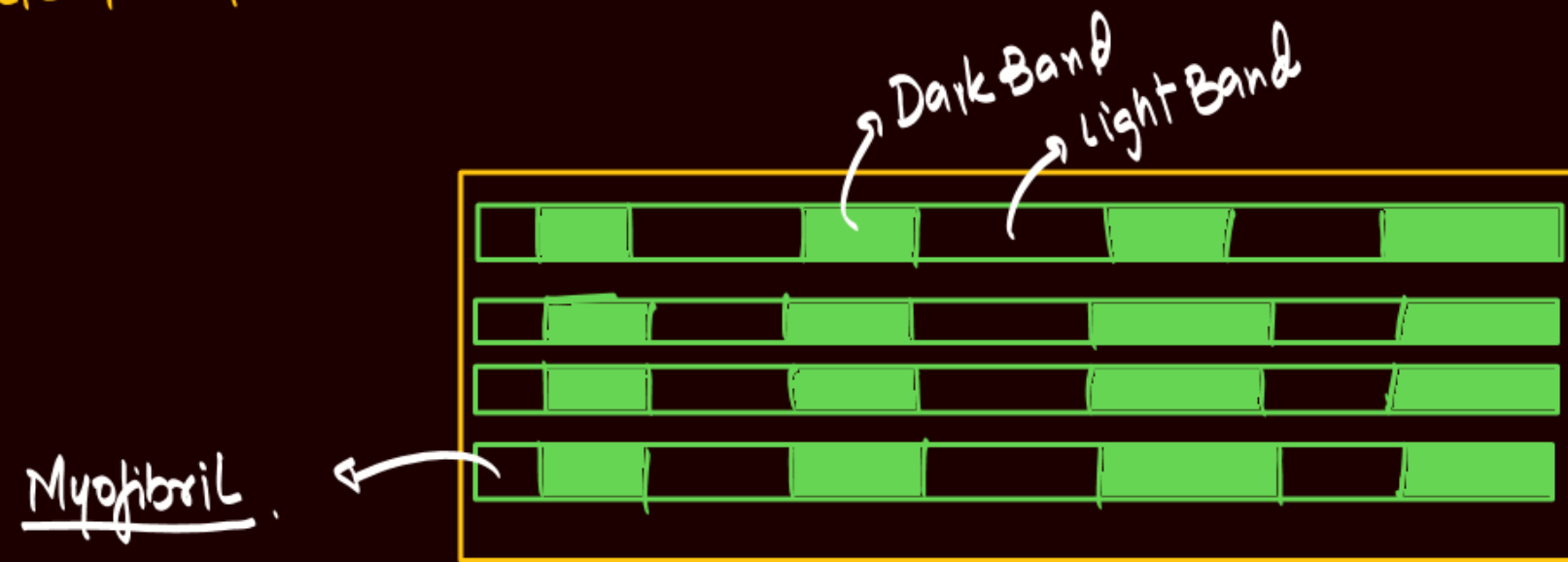
- Ca^{++} Stores in = ③

Skeletal Muscle



→ Muscle Fibre - Continue

→ Muscle fibre / cell contains Myofibril.



→ Myofibril are elongated structures, made of Myofilaments (contractile proteins)

→ Myofilaments are of two types ; Thin Filament (Actin), thick Filament (Myosin).

→ The arrangement of thin & thick Filament forms Alternate Light & Dark Bands in Myofibril.

Arrangement of Actin & Myosin Filament In a Myofibril

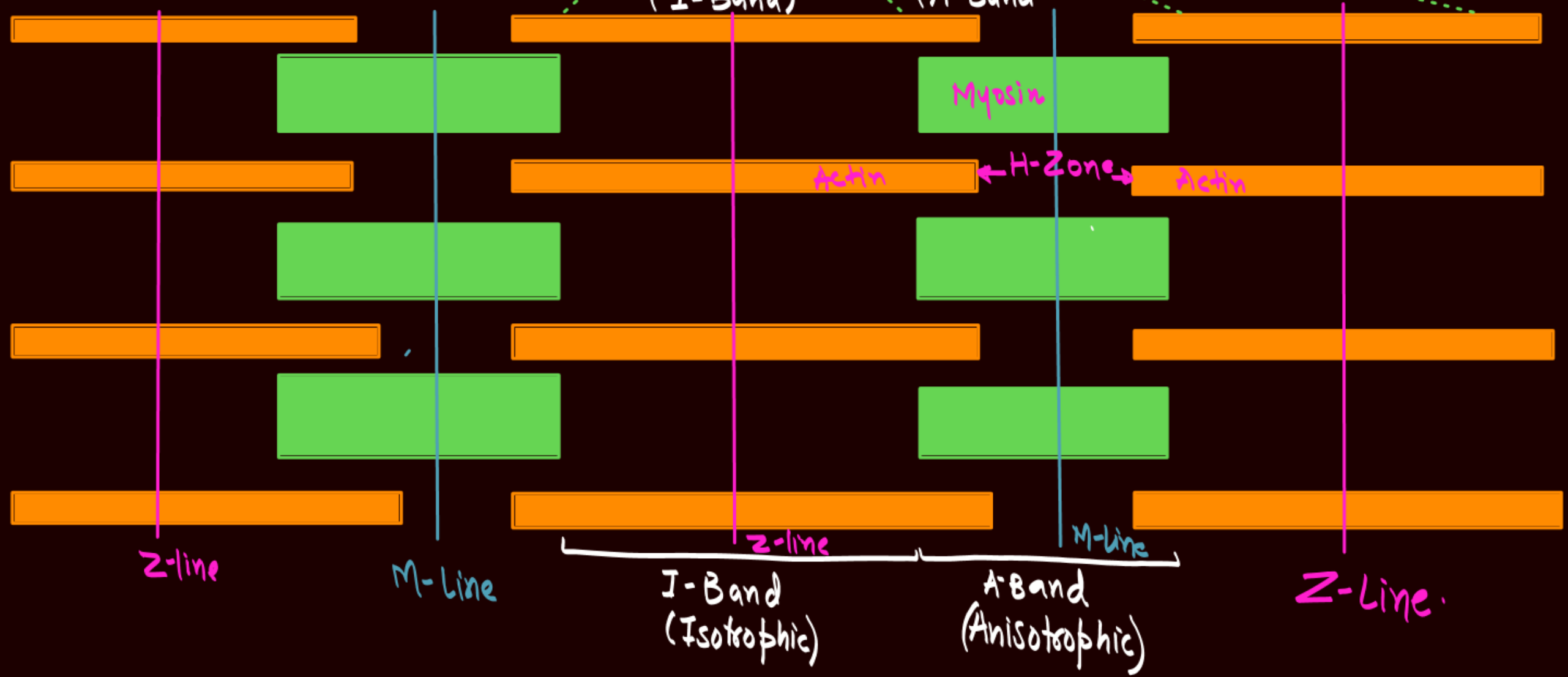
↳ thin Filament partially overlaps thick Filament at Rest

Actin Filament = 

Myosin Filament = 



Light Band (I-Band)
Dark Band (A-Band)
Light Band



Z-line

M-Line

I-Band
(Isotrophic)

A-Band
(Anisotropic)

Z-Line

Sarcomere

Sarcomere = Unit of Muscle Contraction

→ Z-line - it passes through centre of I-Band.
- it is Elastic.

→ M-line - it passes through centre of A-Band.
- it is Fibrous.

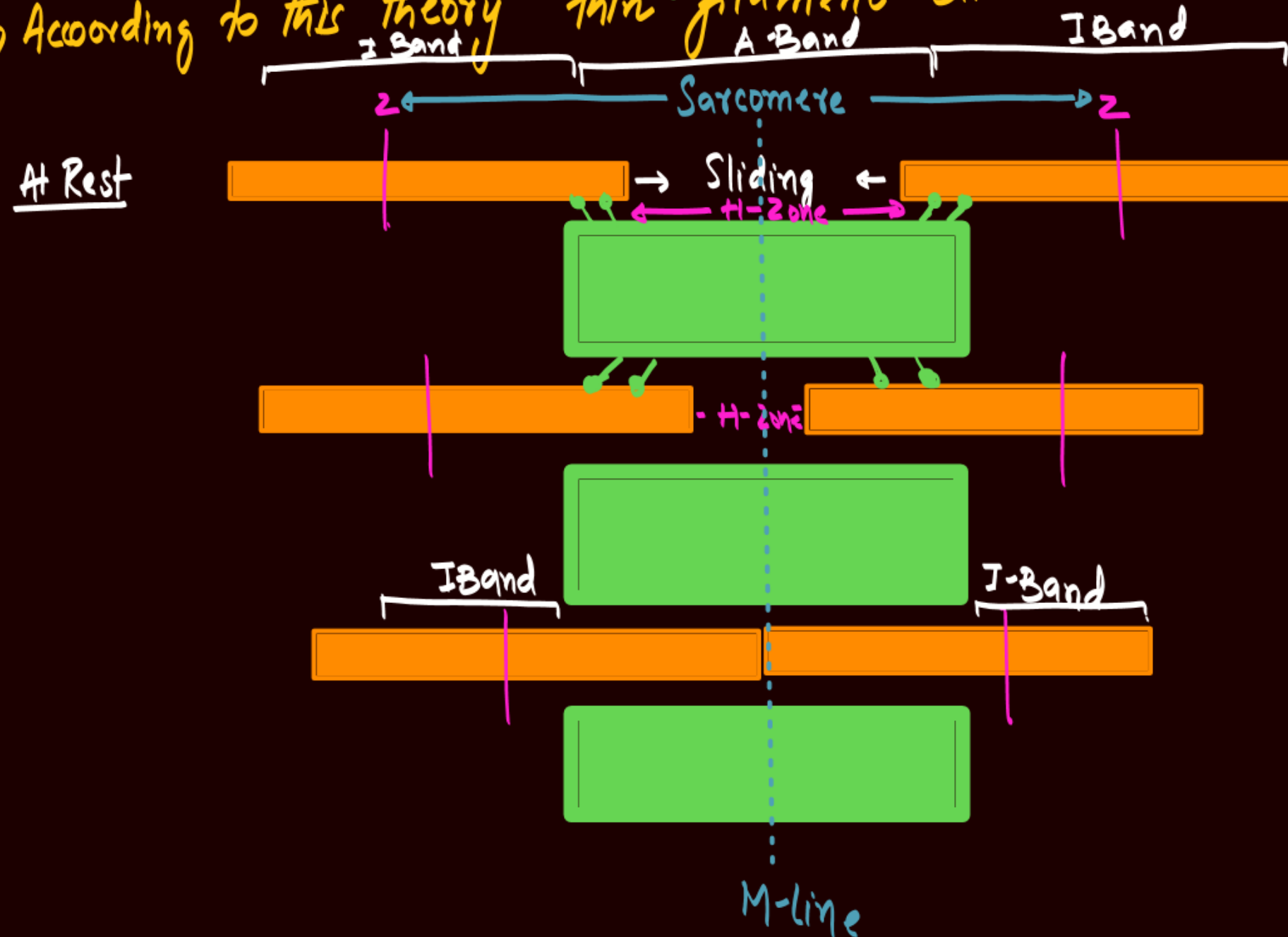
→ H-Zone - It is non-overlapped zone on A-Band.

. That zone where Actin has not overlapped.
at Rest.

Theory of Muscle Contraction

↳ Muscle contraction is explained by Sliding Filament theory. (By Huxley & Huxley)

↳ According to this theory "thin filament slides on thick filament"



- Changes at Contraction

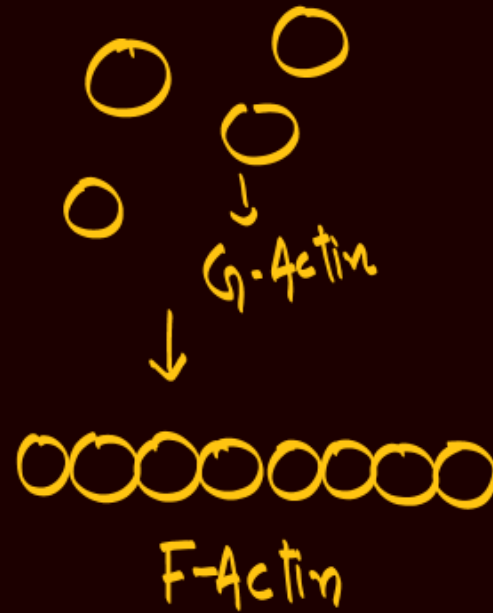
- ① Z-line come closer (contraction)
- Length of Sarcomere Decrease
- ② A-Band Remains Same
I-Band Decreases
- ③ H-Zone decrease / Disappears
- ④ M-line Remains Same.

Structure of Actin

↳ Actin is made of Z-F Actin, Tropomyosin, Troponin.

① ↳ F-Actin - (Filamentous Actin)

- Made of G-Actin (G = Globular)
- Helically present



② ↳ Tropomyosin is a protein chain which runs parallel to F-Actins

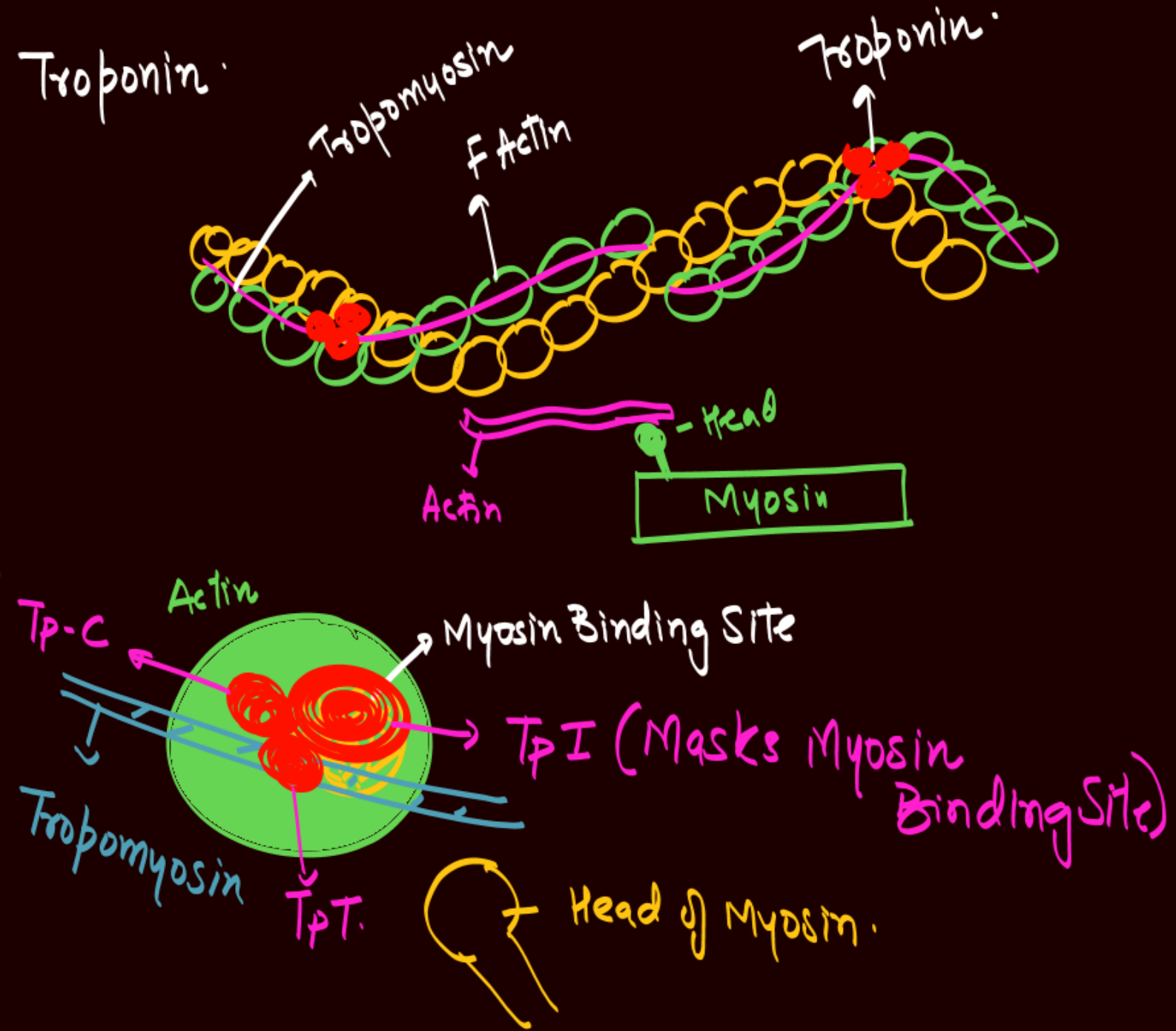
③ ↳ Troponin

- These are complex protein present at places.

- Having 3-subunits ① TpI → Attached to Actin
↳ It inhibits joining of Actin with Myosin, by Masking Myosin Binding site.

② TpT → Attached to Tropomyosin.

③ TpC → to it Ca^{++} Binds.



→ Binding of Ca^{++} with Tp-C, causes ^(conformation) Shape change of Troponin, leading to
unmasking of Myosin Binding Site

Thank
You